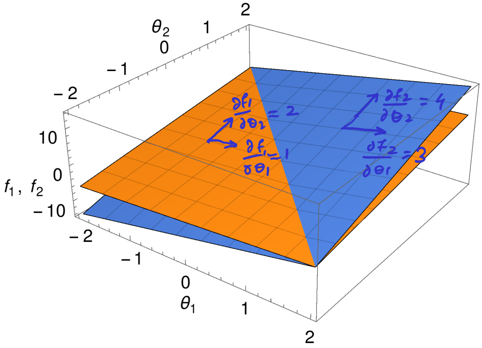
Derivative of a linear transformation $\mathbf{f} : \mathbb{R}^2 \to \mathbb{R}^2$

$\mathbb{R}^n \to \mathbb{R}^m$



$$\mathbf{f}(\boldsymbol{\theta}) = \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}}_{X} \underbrace{\begin{pmatrix} \theta_1 \\ \theta_2 \end{pmatrix}}_{\boldsymbol{\theta}} = \begin{pmatrix} \theta_1 + 2\theta_2 \\ 3\theta_1 + 4\theta_2 \end{pmatrix}$$

column

$$\mathbf{f}'(\boldsymbol{\theta}) := \left(\left(\frac{\partial f_j}{\partial \theta_i}\right)\right) = \begin{pmatrix} \frac{\partial f_1}{\partial \theta_1} & \frac{\partial f_2}{\partial \theta_1} \\ \frac{\partial f_1}{\partial \theta_2} & \frac{\partial f_2}{\partial \theta_2} \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = X^\intercal$$

row

W. Rudin, *Principles of mathematical analysis* (McGraw-Hill, 1976), Chapt 9.

Derivative of a quadratic form $f : \mathbb{R}^n \to \mathbb{R}$

$n=2$

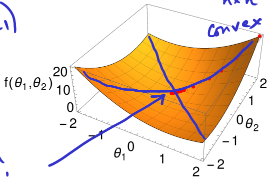
$$f(\boldsymbol{\theta}) = \underbrace{\begin{pmatrix}\theta_1 & \theta_2\end{pmatrix}}_{\boldsymbol{\theta}^{\mathsf{T}}} \underbrace{\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}}_{X_{n\times n}} \underbrace{\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}} = 2\left[\theta_1^2+\theta_2^2+\theta_1\theta_2\right]$$

$3,\ 1 > 0$

$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}1\\-1\end{pmatrix}$

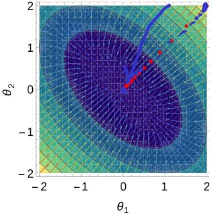
convex

global min



$$\frac{\partial f}{\partial \theta_1} = 2\left[2\theta_1+\theta_2\right]$$

$$\frac{\partial f}{\partial \theta_2} = 2\left[\theta_1+2\theta_2\right]$$



$H = 2X > 0$

Positive def

$$\nabla f = 2 \underbrace{\begin{pmatrix}2 & 1\\ 1 & 2\end{pmatrix}}_{X} \underbrace{\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}}$$

$$f(\boldsymbol{\theta}) = \underbrace{\begin{pmatrix}\theta_1 & \theta_2\end{pmatrix}}_{\boldsymbol{\theta}^{\mathsf{T}}} \underbrace{\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}}_{X} \underbrace{\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}}$$

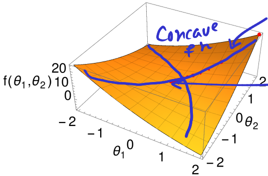
$3,\ -1$

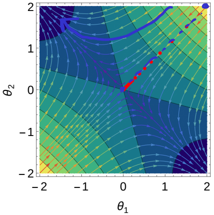
$\begin{pmatrix}1\\1\end{pmatrix}$ $\begin{pmatrix}1\\-1\end{pmatrix}$

Concave fn

Convex

Saddle Point





$H = 2X$

Indefinite

$$\nabla f = 2 \underbrace{\begin{pmatrix}1 & 2\\ 2 & 1\end{pmatrix}}_{X} \underbrace{\begin{pmatrix}\theta_1\\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}}$$

Derivative

① Linear Tran $f = X\theta$ $\quad f' = X^T$

$X = X^T \Rightarrow$ Symm Mat

② **quadratic form** $f : \mathbb{R}^n \to \mathbb{R}$

$(A+B)^T = A^T + B^T$

$(AB)^T = B^T A^T$

$$f(\boldsymbol{\theta}) = \underbrace{\begin{pmatrix}\theta_1 & \theta_2\end{pmatrix}}_{\boldsymbol{\theta}^\intercal} \underbrace{\begin{pmatrix}x_{11} & x_{12} \\ x_{21} & x_{22}\end{pmatrix}}_{X_{n\times n}} \underbrace{\begin{pmatrix}\theta_1 \\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}} \qquad \nabla f = \underbrace{\left[\underbrace{\begin{pmatrix}x_{11} & x_{12} \\ x_{21} & x_{22}\end{pmatrix}}_{X} + \underbrace{\begin{pmatrix}x_{11} & x_{21} \\ x_{12} & x_{22}\end{pmatrix}}_{X^\intercal}\right]}_{2X} \underbrace{\begin{pmatrix}\theta_1 \\ \theta_2\end{pmatrix}}_{\boldsymbol{\theta}}$$

$X_{m\times n}$

$\theta_{n\times 1}$

$y_{m\times 1}$

$$\text{loss} = f(\boldsymbol{\theta}) = [X\boldsymbol{\theta} - y]^\intercal [X\boldsymbol{\theta} - y] = \underbrace{\overbrace{\boldsymbol{\theta}^\intercal X^\intercal}^{(X\theta)^T} X\boldsymbol{\theta}}_{\text{Symm}} \underbrace{- y^\intercal X\boldsymbol{\theta} - \underbrace{\overbrace{\boldsymbol{\theta}^\intercal X^\intercal}^{(X\theta)^T} y}_{y^\intercal X\boldsymbol{\theta}}}_{-2(y^Tx)\theta} + y^\intercal y$$

$$\nabla f = 2\left(X^\intercal X\boldsymbol{\theta} - X^\intercal y\right)$$

$f : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$

$$f(\boldsymbol{\theta}) = \underbrace{\begin{pmatrix}x_1 & x_2\end{pmatrix}}_{X^\intercal} \underbrace{\begin{pmatrix}\theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22}\end{pmatrix}}_{\boldsymbol{\theta}} \underbrace{\begin{pmatrix}x_1 \\ x_2\end{pmatrix}}_{X} \qquad \nabla f := \left(\left(\frac{df}{d\theta_{ij}}\right)\right) = \underbrace{\begin{pmatrix}x_1 \\ x_2\end{pmatrix}}_{X} \underbrace{\begin{pmatrix}x_1 & x_2\end{pmatrix}}_{X^\intercal}$$

row column

$$= \sum_{ij} x_i \theta_{ij} x_j = x_i x_j$$